PAPER_I_EXT_AUDIT — 03_split_falsification

Script 03 — Split-Graph Falsification of Theorem 1.1

Audit: PAPER_I_EXT_AUDIT · **Script:** `03_split_falsification.py`

Environment: Python 3.14.4, scipy 1.17.1 (HiGHS LP), numpy · seed 20260710

Purpose

Directly test Theorem 1.1 — $|E(G)| - 2\nu\blacksquare^*(G) \le n^2/6 + n$ — on split graphs, computing $\nu\blacksquare^*(G)$ **exactly** by solving the fractional triangle-packing LP on the whole graph (independent of the manuscript's two-phase construction). Also confirm the extremal family attains the leading constant.

Split graph model: clique K_p plus independent vertices with neighborhoods $N(v) \subseteq K$. Triangles are clique triples and mixed triples $\{v,x,y\}$ with $x,y \in N(v)$ (no triangle uses two independent vertices, since I is independent).

Predeclared kill-switches

- **KS1** any split graph with $|E| - 2\nu\blacksquare^* > n^2/6 + n + 1e{-}6$ ■ **THEOREM FALSIFIED**.
- **KS3** extremal $K\_p \vee K\blacksquare\_\{2p\}$ with $|E| - 2\nu\blacksquare^* \neq n^2/6 + n/6$ ■ tightness claim wrong.

Instances and results

- **(a)** structured enumeration $p=2..5$, one independent vertex per subset of K (all neighborhood types): all satisfy the bound (ratios 0.25–0.30).
- **(b)** 600 random split graphs ($p=2..7$, up to 10 independent vertices, random neighborhoods): all satisfy the bound.
- **(c)** adversarial complete-split $N(v)=K$ with up to $3p$ independent vertices: all satisfy the bound.
- **(d)** extremal family $K\_p \vee K\blacksquare\_\{2p\}$ (n=3p), $p=2..7$: $\nu\blacksquare^* = C(p,2)$ exactly, and $|E| - 2\nu\blacksquare^* = n^2/6 + n/6$ exactly on every instance.

Quantity	Value
total split graphs tested (a,b,c)	622
theorem violations (KS1)	**0**
max ratio $(|E|-2\nu\blacksquare^*)/(n^2/6+n)$	0.815 (at K■∨K■■■)
extremal tightness failures (KS3)	**0**

Verdict

NOT FALSIFIED. No split graph violated $|E| - 2v■* \le n^2/6 + n$; the extremal family $K_p \vee K■_{2p}$ attains $n^2/6 + n/6$ exactly, matching the manuscript's §9 discussion (the leading constant $1/6$ is tight; the additive $+n$ is not optimized — observed peak $n/6$). **Limit:** a bounded computational search is supporting/typed evidence for a universal (∀) claim, never a proof; the analytic proof (Scripts 01–02 + Gate H) carries the universal statement.